

Day 18 – Model / Method Comparison & Mentor Review

Battery Health SoH & RUL Prediction using Deep Learning

Dataset (V1)

- Dataset V1 — 2,287 sequences
- Sequence length = 10 cycles
- 2 input features
- Targets = SoH + RUL (simultaneous prediction)
- Sequence-based deep learning for all models

Data Split

Split	Samples
Train	1,516
Validation	379
Test	392

Total = 1,516 + 379 + 392

Models Developed

Model	Architecture	Key Configuration
LSTM	2-layer LSTM	Hidden size 64
GRU	2-layer GRU	Hidden size 64
TCN	2 TCN blocks	Channel sizes 32 → 64

Evaluation Metrics

- MAE
- RMSE
- MAPE
- R²
- Training time
- Inference time
- Model size / parameter count

Fair-Comparison Protocol

- Same Dataset V1 and same train / validation / test split used for LSTM, GRU and TCN.

“Identical dataset split and evaluation procedure were maintained across all three architectures for fair comparison.”

LSTM vs GRU vs TCN — Performance Comparison

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Model	Target	MAE	RMSE	MAPE	R ²	Training Time	Inference Time	Parameters
LSTM	SoH	21.813	24.004	$3.42 \times 10^{17} \%$	-1.415	Not Recorded	0.0183 s	50,818
LSTM	RUL	2.161	5.491	$9.39 \times 10^{17} \%$	-57.889	Not Recorded	0.0183 s	50,818
GRU	SoH	9.852	15.600	$4.78 \times 10^{17} \%$	-0.020	82.66 s	0.2420 s	38,146
GRU	RUL	9.804	9.830	$4.37 \times 10^{18} \%$	-187.725	82.66 s	0.2420 s	38,146
TCN	SoH	12.371	17.157	$4.12 \times 10^{17} \%$	-0.234	56.21 s	0.0648 s	24,226
TCN	RUL	2.912	3.152	$1.24 \times 10^{18} \%$	-18.409	56.21 s	0.0648 s	24,226

Best value per target among the three models (MAE, RMSE, R²) and best recorded training time / parameter count. All R² values are negative.

Note: MAPE produced extremely large values and was therefore treated cautiously during model comparison. Greater emphasis was placed on **MAE, RMSE and R²**.

Key Observations

- **GRU** achieved the best **SoH** performance:
 - MAE = 9.852
 - RMSE = 15.600
 - R² = -0.020
- **LSTM** achieved the lowest **RUL MAE**:
 - MAE = 2.161
- **TCN** achieved the best **RUL RMSE**:
 - RMSE = 3.152
 - R² = -18.409, the best RUL R² among the three models
- **TCN** has the smallest model size: **24,226** parameters
- **TCN** has the fastest recorded training time: **56.21 s**
- **TCN** inference time: **0.0648 s**

1 • Overfitting / Validation Behaviour

- **LSTM:** best val. epoch = 92, best val. loss = 296.11. Validation performance became unstable / fluctuated in later training.
- **GRU:** best val. epoch = 35, best val. loss = 340.86. Validation loss plateaued / increased after the best epoch while training loss kept improving.
- **TCN:** best val. epoch = 14, best val. loss = 262.23. Training loss kept decreasing while validation loss increased, indicating overfitting. **Early checkpoint selection is important.**

2 • Limitations / Failure Patterns

- **All three models produced negative R^2** for both SoH and RUL.
- Current models are not yet achieving satisfactory performance compared with a simple baseline.
- MAPE values are extremely large; not used as the primary selection criterion.
- Dataset V1: only 2 input features and 10-cycle sequences may limit predictive capability.
- Additional feature engineering, longer temporal context, or improved target / loss handling may be required.

3 • Shortlisted Candidates

Primary: TCN

- Lowest parameter count: 24,226
- Strong RUL RMSE: 3.152; best RUL R^2 of the three: -18.409
- Fastest recorded training time: 56.21 s; inference 0.0648 s
- Compact architecture

Secondary: GRU

- Best SoH MAE 9.852, RMSE 15.600, R^2 -0.020
- Fewer parameters than LSTM: 38,146

LSTM: baseline / reference

- Lowest RUL MAE = 2.161, but larger parameter count and poor R^2 reduce priority for deeper experiments

4 • Next Direction

Proceed with controlled deeper experiments on the shortlisted TCN and GRU architectures.

- Improve input feature representation
- Investigate longer sequence lengths
- Address validation overfitting
- Investigate loss weighting for SoH and RUL
- Reassess metrics after improving prediction quality
- Maintain the same test protocol for fair comparison

Conclusion. Based on the current controlled comparison, TCN is shortlisted as the primary candidate for deeper RUL experiments, while GRU is retained as a strong candidate for SoH prediction. The negative R^2 values and unstable MAPE indicate that further feature, sequence, and training strategy improvements are required before final model selection.